NYAYMITRA 2.0 — Prototype Progress and Evidence Report

Implemented, tested, demonstrated, incomplete, test outputs, commit references

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1. What has been implemented

- Backend: FastAPI app factory, config, security, middleware, logging (PII redaction), models, routers, services
- AI/RAG: provider abstraction (openai_compatible + offline), retrieval, citation validation, safety guard, registry
- Frontend: citizen SPA + admin console (English/Gujarati, accessible)
- Database: SQLite, 4 migrations, seed data (9 articles, 11 departments, 5 triage guides, 18 glossary terms), 2,125 quarantined legacy rows
- Scripts: init_db, check_db, backup_db, purge_data, ingest_constitution, constitution_coverage, license_audit, verify_ai_provider, verify_changeset, smoke_web
- Deployment: Dockerfile, systemd unit, render.yaml, .env.example
- Documentation: security, privacy, IP, legal, setup, audit, architecture, API, schema, testing

2. What has been tested

`pytest` full suite	226 passed, 0 failed
Coverage `--cov=backend/app`	92% (2,545 statements, 202 missed)
`pip-audit` dependencies	No known vulnerabilities
`scripts/check_db.py`	OK (integrity ok, 4 migrations, 0 missing indexes)
`node --check` frontend	Pass
`scripts/verify_ai_provider.py`	Exit 3 — no model runtime (BLOCKED)
jsdom smoke test (live server)	All citizen journeys + admin console OK
Frontend static checks	Pass

3. What has been demonstrated

Against a live `uvicorn` server on `0.0.0.0:8000`:

- Home, assistant (answer + citations + disclaimer), library search, glossary filter
- Language toggle EN ↔ GU persists
- Complaint: preview → consent checkbox → submission → reference + access token (NYM-XXXXXX)
- Saved-complaint lookup by reference + token; wrong token → generic 404 (no oracle)
- Admin console: stats, verification queue, metrics, documents
- All routes return expected status; CSP/security headers present

4. What remains incomplete

Official Constitution source documents	Not loaded	B-1: owner must supply PDFs
Human legal review of articles/departments/guides	0 of 9 verified	B-2: reviewer not appointed
Real AI model inference	Not available	B-4: no Ollama/model runtime
Real-device / screen-reader testing	Not done	B-3: no browser/device in sandbox
Public deployment	Not deployed	B-5: owner approval required

5. Test outputs

``` 226 passed, 1 warning in ~20s TOTAL 2545 statements 202 missed 92% pip-audit → No known vulnerabilities found PRAGMA integrity_check = ok; 4 migrations applied, 0 missing indexes /scripts/verify_ai_provider.py → exit 3 (no provider configured) ```

6. GitHub commit references

- Repository: https://github.com/dhundhwana-arch/Nyaymitra-2.0
- Branch: `arena/812d6894-nyaymitra-2-0`
- Audited source snapshot: `cc56d0e` (tests and validation executed here)
- Source snapshot used for PDF generation: `cc56d0e`
- Commit on which tests were executed: `792b529` (226 passed)
- CI run on source snapshot: `42595255121` (push, success)
- Prior commit: `792b529` (source validation commit)
- Earlier: `5a6dcb9` (AI reliability, usage tracking, review packet, deployment configs)
- PR #1: https://github.com/dhundhwana-arch/Nyaymitra-2.0/pull/1 (open, draft, mergeable)

7. Screenshots

No screenshots are included. Behaviour was verified against a live server via the `scripts/smoke_web.mjs` jsdom test; all journey output is recorded above. No fabricated or invented screenshots are used.

8. Repository

https://github.com/dhundhwana-arch/Nyaymitra-2.0

NYAYMITRA 2.0 — functional software prototype / Proof of Concept — subject to independent evaluation. Not legal advice; not a replacement for courts, lawyers, police, or government authorities. https://github.com/dhundhwana-arch/Nyaymitra-2.0 (branch arena/812d6894-nyaymitra-2-0, commit cc56d0e)
```